



PETROVA

Production-Grade Data Platform

Offshore asset monitoring · Snowflake · Databricks · dbt · Airflow · synthetic data

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github.com/Kepler22j/petrova-pipeline · petrova-pipeline.streamlit.app



A pipeline that fails silently is dangerous

PETROVA moves ~400,000 equipment-sensor records every day.
If a load fails quietly, two things break:

Data loss / stale dashboards

The business makes decisions on old, unvalidated data — without knowing it.

Missed equipment alarms

A failing asset (e.g. a compressor bearing) goes undetected until it breaks.

PETROVA makes pipeline health + equipment anomalies visible in one place — caught in minutes, not after the damage.



Medallion lakehouse, dual-cloud



Orchestration

Apache Airflow · Azure Data Factory · Databricks Workflows

Compute & storage

Snowflake (SQL) + Databricks (PySpark / Delta) · ADLS Gen2



Every record passes 3 gates

GATE 1 — BRONZE

Schema & nulls

- Required columns present
- Type validation
- NOT NULL on keys

FAIL → QUARANTINE

GATE 2 — SILVER

Business rules

- Range / status checks
- Dedup (ROW_NUMBER)
- SCD2 + referential integrity

FAIL → QUARANTINE

GATE 3 — GOLD

SPC / statistical

- Anomaly severity check
- Blocks on CRITICAL
- FMEA risk gate

FAIL → QUARANTINE



SPC engine: 6 alerts from 3 primitives

STDDEV

spread

LAG

change

THRESHOLD

limits

▼ derive ▼

1. Stability level

2. Signal quality (CV)

3. Outlier (2σ)

4. Range violation

5. Volatility trend

6. Spike (LAG)

**COMPOSITE
SEVERITY**

 **CRITICAL**

 **WARNING**

 **OK**

UNSTABLE + noisy → CRITICAL



The engine catches a real fault

493

OK

40

WARNING

7

CRITICAL

Injected a bearing-degradation fault on the K-201 gas compressor: vibration stddev ramps up over ~3 weeks, crosses the control limit, and the SPC engine escalates it to CRITICAL — proving early-warning detection on a known failure mode.

► Live: petrova-pipeline.streamlit.app · run: `py scripts/send_alert.py`



Dashboards never go dark

Two layers keep BI available even when a rebuild fails:

Layer A — trigger_rule = all_done

Post-processing runs even on partial failure; bad records route to quarantine, not Gold.

Layer B — Zero-Copy Clone cache

A snapshot of Gold is cloned before each rebuild — instant, no storage cost. If the rebuild breaks, BI serves the clone.

Result: data freshness protected; on-call escalation only when it truly matters.



Measured, not claimed

125K

records/sec (benchmarked)

400K+

records/day (configurable)

16

dbt models · 3 layers

99.5%

SLA at PTTEP (production)

Synthetic data by design (no employer data) · config-as-code (env-driven dbt profiles) · fault-injection testing



Real experience + safe demonstration

PTTEP — Production

- Real APM platform, 6 offshore rigs
- 400K+ records/day · 99.5% SLA
- SAP/PDMS → APM alerting
- Confidential — described, never shown

PETROVA — Portfolio

- Modern stack rebuild of those patterns
- Snowflake · Databricks · dbt · Airflow
- Synthetic data — reproducible & shareable
- Public: GitHub + live dashboard

"The data is synthetic; the engineering is real."



Let's talk.

Senior Data Engineer · ETL · Analytics — production rigor, modern stack, honest engineering.

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